

Quasi-Experimental Designs

Method Dossier with Field-Specific Examples | Standard Level

1. Method Description

Quasi-experimental designs address causal inference when randomization is infeasible. Four key approaches:

Difference-in-Differences (DiD)

Compares change in treatment group before/after intervention vs change in control. Assumption: parallel trends. $DiD = (Y_{T_post} - Y_{T_pre}) - (Y_{C_post} - Y_{C_pre})$.

Regression Discontinuity (RDD)

Assignment by cutoff on running variable. Causal effect = discontinuity at cutoff. Assumption: continuity of conditional expectations, no manipulation.

Instrumental Variables (IV)

Z affects treatment X but not outcome Y (exclusion restriction). $\beta_{IV} = Cov(Z,Y)/Cov(Z,X)$. Assumptions: relevance, exclusion, monotonicity.

Propensity Score Matching (PSM)

Match treated to controls on propensity score $P(T=1 | X)$. Reduces observable differences.

2. Field Examples

Social Sciences: DiD on minimum wage introduction (treatment state vs neighbor state).

Economics: RDD on scholarship cutoff grade 2.0 -> effect on graduation grade.

Medicine: IV — distance to vaccination center affects vaccination status but not health.

Sports Sciences: DiD on training program in school vs matched control school.

3. Assumptions

- DiD: parallel trends, no asymmetric shocks, stable group composition
- RDD: continuity, no manipulation (McCrary test), appropriate bandwidth
- IV: relevance ($F > 10$), exclusion restriction (untestable), monotonicity
- PSM: conditional independence (no unobserved confounders), common support

4. Interpretation and Reporting

- Name design and state identifying assumption explicitly
- Report causal effect with 95% CI
- Balance checks before/after matching
- Sensitivity: placebo tests, different bandwidths/matchers
- Graphics: parallel trends, RDD plot, propensity score density

5. Code Examples

R

```
did::att_gt(y ~ post*treat + id + year)
rdrobust::rdrobust(y ~ x, c=cutoff)
AER::ivreg(y ~ x | z)
MatchIt::matchit(treat ~ x1 + x2)
```

Python

```
linearmodels: PanelOLS, IV2SLS
rdrobust Python wrapper
```

6. Extension

Callaway & Sant'Anna (2021) Generalized DiD

Handles staggered adoption, heterogeneous treatment effects.

Fuzzy RDD

When treatment probability jumps by < 1 at cutoff. Becomes IV design.

Synthetic Control

For single treated unit. Weighted control units match pre-treatment trends.

Causal Forest

ML-based heterogeneous treatment effects (CATE).

7. Exercises

Exercise 1: State introduces universal basic income in 2020. Sketch DiD: control group? DV? Assumptions tested how?

Exercise 2: Scholarship at grade cutoff 2.5. a) RDD sketch. b) Bandwidth choice? c) Interpret significant jump at cutoff.

Exercise 3: Z (dictionary availability) affects study time X but not exam grade Y directly. Valid instrument? Which assumption potentially violated?